

The pubertal timing controversy in the USA, and a review of possible causative factors for the advance in timing of onset of puberty.

Previously used standards for the diagnosis of precocious puberty in girls no longer appear to be appropriate in the USA, in that a significant number of girls are being seen in paediatricians' offices with breast budding before 8 years of age. The timing of menarche, however, has changed little over the past few decades. Early maturing girls are more likely to become obese in adolescence and adulthood than normal or late maturing girls. Early maturing white girls are heavier at the onset of puberty, but this is not the case for African-American girls or boys of either race. Boys and girls with premature pubarche may be more hyperinsulinaemic than normal children, and girls with premature pubarche more likely to develop functional ovarian and adrenal hyperandrogenism. Early menarche is preceded by prepubertal hyperinsulinaemia. It is proposed that pubertal onset, although not necessarily the tempo of puberty, is influenced by hyperinsulinaemia and insulin resistance. If this hypothesis is correct, insulin resistance may be more prevalent in US children than previously recognized. An advance in timing of onset of puberty has not been noted in other countries, although it is likely that this phenomenon may become more prevalent as other countries adopt a more American lifestyle and diet.